DESIGN COMPETITION ENTRY • ASSISTIVE IOT TECHNOLOGY

AegisRadar: Privacy-First Elderly Care Monitoring

Non-invasive 24GHz mmWave radar activity tracking and automated behavioral anomaly detection for safe independent living.

Zero Cameras or Microphones

LD2410c FMCW Radar

PROBLEM STATEMENT

Balancing Safety with Personal Dignity

Privacy Intrusiveness

Traditional optical cameras cause discomfort and violate elderly privacy in bathrooms and bedrooms.

Undetected Emergencies

Falls or medical incidents inside private rooms often go unnoticed for hours without active monitoring.



TECHNICAL ARCHITECTURE

End-to-End Non-Invasive Data Pipeline

1. LD2410c Radar

24GHz FMCW sensing detects static breathing and motion without visual lenses.

2. Edge Node

Microcontroller processes raw UART presence signals and streams room states.

3. Central Server

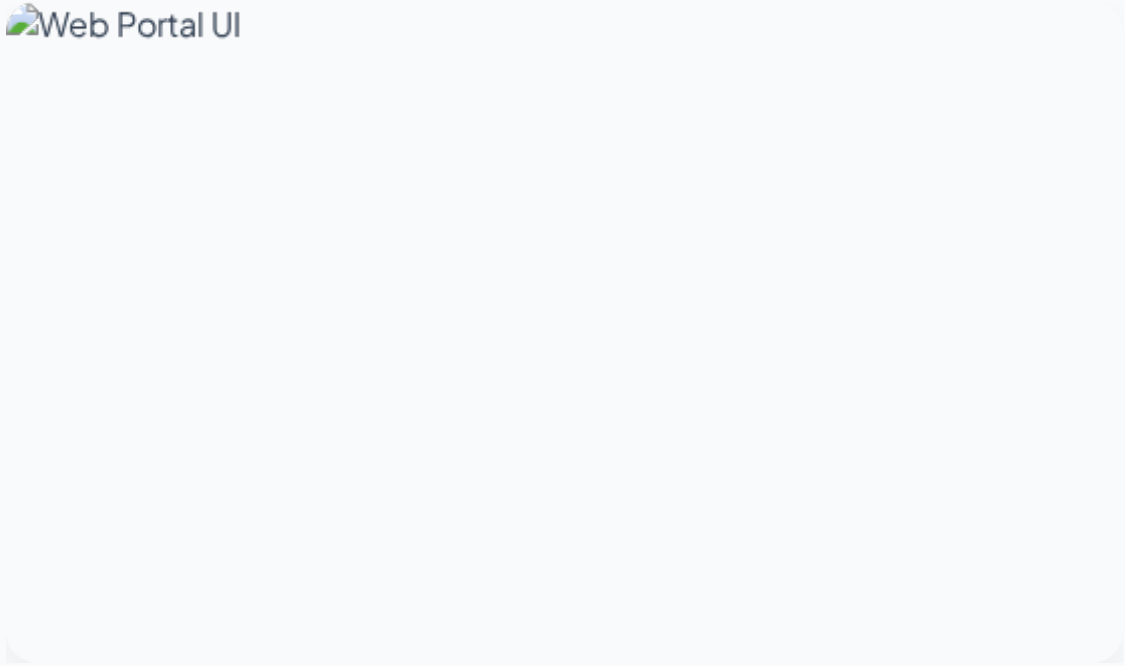
Logs movement timestamps, aggregates room metrics, and evaluates safety rules.

4. Web Portal

Displays real-time occupancy heatmaps and dispatches emergency alerts.

CENTRAL DASHBOARD

Real-Time Activity Verification & Timestamps



Instant Room Location

Family members verify exact room presence at a single glance.

Heartbeat Timestamps

Continuous telemetry timestamps confirm node operational health.

Passive Assurance

Daytime movement confirms wellness without intrusive check-in calls.

INTELLIGENT ANALYTICS

Bathroom & Bedroom Safety Rule Engine

Duration Tracking

Records exact entry and exit timestamps. Flags stay times exceeding custom safety thresholds.

Frequency Analysis

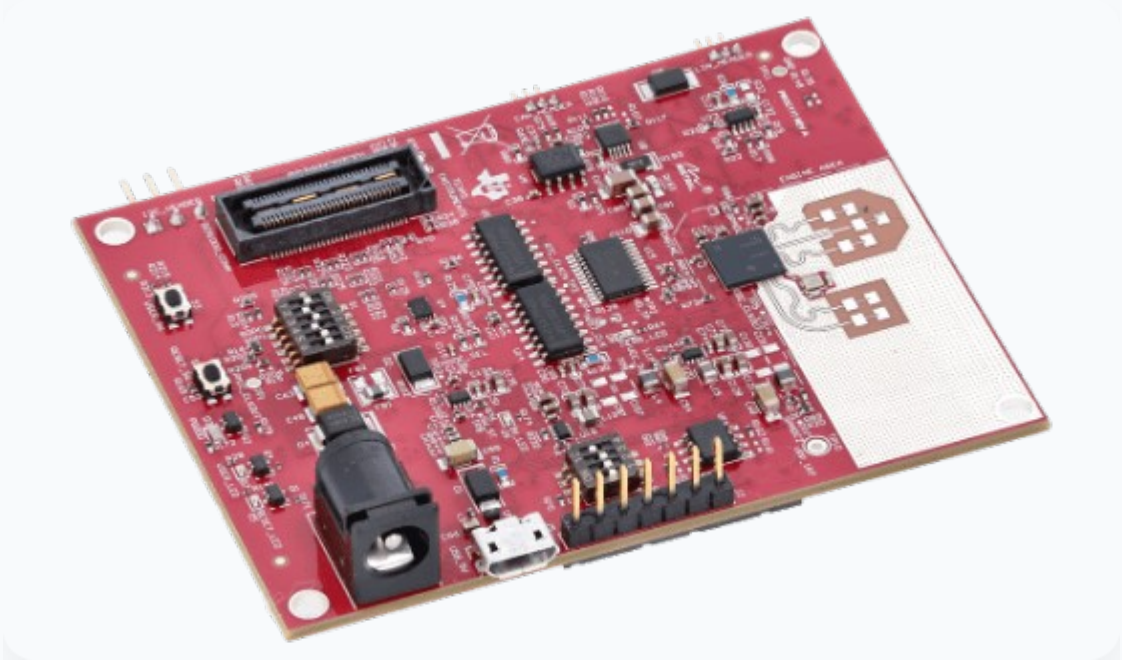
Monitors day vs. night visit frequency to identify health indicators like sleep disruption or infection.

Automated Alerts

Triggers instant SMS or dashboard alerts when abnormal room retention implies a potential fall.

SYSTEM PROTOTYPE

Hardware Implementation & Real Demonstration



Low-Cost Radar Sensors

LD2410c module senses micro-movement chest expansions up to 5 meters.

Robust Microcontroller Node

Compact wireless nodes deployed across key rooms for full home coverage.

Validated Live Demo

Successfully recorded functional test showing automated alert triggering.

NEXT STEPS

Future Development Roadmap

Phase 1

Machine Learning Routine Profiling

Automatically baseline individual routine patterns without manual threshold setting.

Phase 2

Multi-Target & Fall Radar Detection

Upgrade to multi-point radar to distinguish multiple occupants and detect sudden height drops.

Phase 3

Caregiver Mobile Application

Native iOS/Android companion app with push notifications and emergency escalation.

Phase 4

Clinical EHR Integration

Export mobility trends to healthcare providers for early detection of cognitive or physical decline.

EMPOWERING INDEPENDENT SENIOR LIVING

Dignity, Safety, and Total Peace of Mind

AegisRadar proves that effective elder care technology can protect lives without invading personal privacy.

Thank You for Your Consideration

Image Sources



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